A Appendix: The Graduate Qualifying Examination in Mathematics (GQE)

A.1 General Comments

The GQE is set by an examining committee appointed by the Department Chair. The Chair of the Graduate Studies Committee, or his/her representative, administers the examination. A unique ID number is assigned to each examinee and only these ID numbers identify the examination papers. The names of the students will be correlated with the examination papers only after the final outcome of the examination has been determined.

Graded copies of the examination papers will be made available in the department office for inspec-tion by the graduate faculty. Each paper will be marked with a consensus score for each problem, a consensus total score, appropriate comments, and the recommended outcome of the examination. After the graduate faculty has had at least a day to weigh the recommendations of the examining committee, the Graduate Faculty will convene to decide the final outcome of the examination for all participating students. It is inappropriate for members of the examining committee or other fac-ulty members to discuss the performance of an examinee with any student, including the examinee, prior to the final announcement of the examination results.

A.2 Topics for the Graduate Qualifying Examination

Here are listed topics that a student should know for passing the GQE. Courses at WSU cover most, if not all, of these topics, but this list is not intended as a syllabus for any course. It is the student's responsibility to prepare adequately.

Topics in Advanced Calculus: Metric spaces, Cauchy sequences and completeness, limits, con-tinuity, Intermediate Value Theorem, differentiation, Mean Value Theorem, Taylor's Theorem, L'Hospital's Rule, monotone functions, convexity and curvature, infinite series, sequences and se-ries of functions, Fundamental Theorem of Calculus, Mean Value Theorem for integrals, improper integrals, partial and directional derivatives, Jacobian matrix, gradient vector, chain rule, Diver-gence Theorem, Stokes' Theorem, Inverse Function Theorem, Implicit Function Theorem, change of variables in multiple integrals, Lagrange multipliers.

References: This material is covered at the appropriate level, for example, in the following texts: T. Kaplan, Advanced Calculus, Addison-Wesley, 2002; A. Taylor and W. R. Mann, Advanced Calculus, Wiley, 1983; W. A. J. Kosmala, A Friendly Introduction to Analysis—Single and Multivariable, Pearson Prentice Hall, 2004.

Topics in Linear Algebra: Vector spaces, subspaces, linear independence, bases and dimen-sion, inner product spaces, norms, triangle inequality, Cauchy-Schwarz inequality, orthogonality, orthonormal bases, orthogonal projections, basic matrix operations, matrix transpose, trace of a matrix, determinants and their properties, invertibility, eigenvalues and eigenvectors, characteristic polynomials, matrix equivalence, matrix similarity, diagonalizability, linear transformations, matrix representations of linear transformations, range and null space (kernel) of a linear transformation, symmetric and hermitian matrices or operators, unitary matrices, normal matrices.

References: This material is covered at the appropriate level, for example, in S. H. Friedberg, A. J. Insel, and L. E. Spence, Linear Algebra, Pearson Prentice Hall, 2003; P. J. Oliver and C. Shakiban, Applied Linear Algebra, Pearson Prentice Hall, 2006.